

Aegis-24 System Requirements

Hardware Requirements

Minimum Configuration

Component	Requirement
CPU	4 cores (Intel i5 or equivalent)
RAM	8 GB
Storage	10 GB available space
Network	Broadband internet connection

Recommended Configuration

Component	Requirement
CPU	8 cores (Intel i7 or equivalent)
RAM	16 GB
Storage	20 GB SSD
Network	High-speed internet connection

Production Configuration

Component	Requirement
CPU	16+ cores
RAM	32+ GB
Storage	50+ GB NVMe SSD

Component	Requirement
Network	Low-latency, high-throughput

Software Requirements

Operating System

- **Linux:** Ubuntu 20.04+, Debian 11+, RHEL 8+
- **macOS:** macOS 12+ (Monterey or later)
- **Windows:** Windows 11 with WSL2

Python Environment

- **Python Version:** 3.10 or higher
- **pip:** Version 22.0 or higher
- **Virtual Environment:** venv or conda recommended

Required Dependencies

```
# Core dependencies (see requirements.txt)
asyncio>=3.4
pytest>=7.0.0
pytest-asyncio>=0.21
pyyaml>=6.0
```

Optional Dependencies

Component	Package	Purpose
Arize Phoenix	arize-phoenix	Observability
LangSmith	langsmith	Trace tracking
E2B SDK	e2b	Sandbox execution
Redis	redis-py	Deduplication cache

Component	Package	Purpose
PostgreSQL	psycopg2	Database storage

Docker Requirements (Optional)

If using Docker deployment:

- **Docker:** Version 20.10+
- **Docker Compose:** Version 2.0+
- **Container Registry Access:** Docker Hub or GHCR

Network Requirements

Firewall Rules

Port	Protocol	Direction	Purpose
443	HTTPS	Outbound	API calls (OpenAI, SEC, GitHub)
6006	HTTP	Inbound	Arize Phoenix UI (local)
5432	TCP	Inbound/Outbound	PostgreSQL (if used)
6379	TCP	Inbound/Outbound	Redis (if used)

API Endpoints

The following external services are accessed:

- `https://api.openai.com` - LLM inference
- `https://www.sec.gov` - Financial filings
- `https://api.github.com` - Code repository access
- `https://api.smith.langchain.com` - LangSmith (optional)

Browser Requirements (for UI)

If accessing Phoenix/LangSmith dashboards:

- **Chrome:** Version 100+
- **Firefox:** Version 90+
- **Safari:** Version 15+
- **Edge:** Version 100+

Performance Benchmarks

Expected Throughput

Operation	Target	Notes
Event Ingestion	>1,000 events/sec	With Redis deduplication
Research Query	<30 seconds	3 iteration max
Remediation Task	<10 minutes	Including sandbox testing
Security Validation	<1 second	Per input/output

Latency Targets

Component	P50	P95	P99
Ingestion	10ms	50ms	100ms
Research	5s	15s	30s
Security	100ms	500ms	1s

Scalability Considerations

Horizontal Scaling

- **Ingestion Bus:** Can be sharded by event type

- **Research Graph:** Stateless, can run multiple instances
- **Dev Swarm:** Queue-based, scale workers independently
- **Safety Gateway:** Stateless, load-balance easily

Vertical Scaling

- Increase LLM context window for complex research
- Add more RAM for larger checkpoint databases
- Use faster CPUs for cryptographic operations

Monitoring Requirements

Metrics to Track

- Token usage per module
- API call latency
- Guardrail violation count
- Sandbox execution time
- Checkpoint database size

Alerting Thresholds

Metric	Warning	Critical
PII Leaks	Any	Any
Jailbreak Success	Any	Any
API Error Rate	>5%	>10%
Latency P95	>2x target	>5x target